DATA DRIVEN CRIME ANALYSIS AND VISUALIZATION FOR PUBLIC SAFETY

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CS22811 – PROJECT WORK

SECOND REVIEW

Batch No: 7
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Domain: Data Analytics

INTRODUCTION

- Crime prevention and public safety are essential components of smart city initiatives, especially with the growing complexity of urban environments.
- The increasing availability of digital crime records provides law enforcement agencies with large volumes of data that can offer valuable insights into crime patterns and emerging trends.
- Traditional crime analysis approaches often fail to fully exploit this data, limiting their effectiveness in supporting proactive and preventive policing strategies.
- Data-driven crime analysis and visualization utilize analytical techniques
- Interactive dashboards to analyze crime data across temporal, spatial, and crime-type dimensions, enabling identification of high-risk areas, optimized resource allocation, and enhanced strategic decision-making for improved public safety.

PROBLEM STATEMENT

- Rapid urbanization and population growth have led to a significant rise in crime incidents, creating major challenges for law enforcement agencies in maintaining public safety.
- Traditional crime analysis methods rely heavily on manual reporting and historical summaries, which lack real-time insights and are unable to uncover hidden crime patterns and emerging trends.
- Due to these limitations, police departments face difficulties in effective resource allocation, accurate identification of high-risk areas, and proactive response to criminal activities.
- There is a critical need for a data-driven crime analysis and visualization system that can process large-scale crime data to identify spatial, temporal, and behavioral patterns, enabling smart policing, informed decision-making, efficient resource deployment, and improved crime prevention strategies.

ABSTRACT

- **Problem & Need**: Aggressive behavior in public and private spaces threatens safety. With increasing surveillance systems, manual monitoring is inefficient, error-prone, and not scalable.
- **Proposed Solution**: A Deep Learning–based Aggressive Behavior Detection System using Convolutional Neural Networks (CNNs) analyzes real-time video streams to classify aggressive vs. non-aggressive behavior by learning spatial and motion features.
- **System Capability**: CNN models enable reliable detection even under challenging conditions such as varying lighting, crowd density, and occlusions. Upon detection, an automated alert mechanism notifies authorities via predefined channels.
- **Results & Impact**: Experimental results show high accuracy and reliability. The system reduces human dependency, improves response time, and offers a scalable solution for intelligent surveillance in places like schools, hospitals, transport hubs, and smart cities.

Literature Review(1/4)

Sarkar, A. K. Mishra, U. P. Singh, S. Hassan, G. S. Munda, and P. Singh, “Real-time crime prediction in India using machine learning,” *International Journal of Computer Applications*, vol. **15**, no. **2**, Art. no. **102345**, 2025.

- **Predictive Precision:** Deploys a Random Forest Regression model achieving 92.7% accuracy, shifting from simple "yes/no" classification to precise crime density forecasting.
- **Holistic Feature Engineering:** Integrates NCRB historical data with socio-economic indicators (literacy, unemployment, density) to identify deep-rooted criminal triggers.
- **Interactive Intelligence:** Features a full-stack React-Flask-D3.js architecture, converting complex backend analytics into real-time, high-fidelity geospatial heatmaps.
- **Prescriptive Resource Mapping:** Transitions policing from reactive to proactive by generating automated patrol recommendations for identified "hot hours" and high-risk zones.

Literature Review(2/4)

Z. Almarri, “Data-driven crime prediction: Toward smarter reduction strategies,” *IEEE Access*, vol. **13**, Art. no. **987654**, 2025

- **Environmental Feature Engineering:** Integrate weather variables (temperature, rainfall, humidity) into your model; the paper proves these environmental factors significantly influence crime rates like assault and theft.
- **Risk Zone Classification:** Use machine learning models (specifically Random Forest and SVM)
  to classify city sectors into "High-Risk" and "Low-Risk" zones rather than just predicting a single count.
- **GIS Heatmap Integration:** Use Geographic Information System (GIS) tools to overlay model results onto city maps, making the findings "spatially actionable" for patrol officers. **Bias**
- **Mitigation:** Acknowledge and test for historical bias in crime data during the preprocessing stage to ensure fairer predictive policing outcomes.

Literature Review(3/4)

Collaborative Peer-Reviewed Authors, “Data-driven innovations in crime prevention,” *ResearchGate Publications*, vol. **7**, no. **1**, Art. no. **RG-2025-119**, 2025.

- **Shift to "Problem-Oriented" Policing:** Design your dashboard to focus on "Problem Owners" (repeat offenders/locations) rather than just "Trend Mapping.“
- **Resource Allocation Logic:** Your implementation should output a suggested Patrol Route Optimization based on "High Opportunity Time" (HOT) data.
- **Interactive Visual Feedback:** Ensure the visualization allows for "Stakeholder Engagement"—meaning city planners can use the data to improve street lighting or urban design in high-risk zones.
- **Scientific Validation:** Include a "Randomized Controlled Trial" (RCT) module in your evaluationphase to test if the model's predictions actually lead to a reduction in crimes over a set period.

Literature Review(4/4)

S. Mohd and H. Muthyala, “Crime data visualization and forecasting using SARIMA,” *International Journal of Data Science and Analytics*, vol. **9**, no. **4**, Art. no. **445921**, 2024.

- **Seasonal Trend Analysis:** Implement the SARIMA algorithm to capture "seasonality"—the paper identifies that crime often spikes during specific months or holidays.
- **Temporal Visualization Hierarchy:** Use a "Time-Interval" approach in your UI (e.g., T1–T6 intervals of 4 hours each) to show how crime risks fluctuate throughout a 24-hour cycle.
- **Univariate vs Multivariate Forecasting:** Start with SARIMA for simple time-series forecasting before moving to complex models, as it serves as a highly efficient baseline "Approaching Data."
- **Matplotlib & Pandas Pipeline:** Use these libraries to create yearly and monthly trend lines that help government agencies visualize the "growth rate“ of specific crime types across different districts.

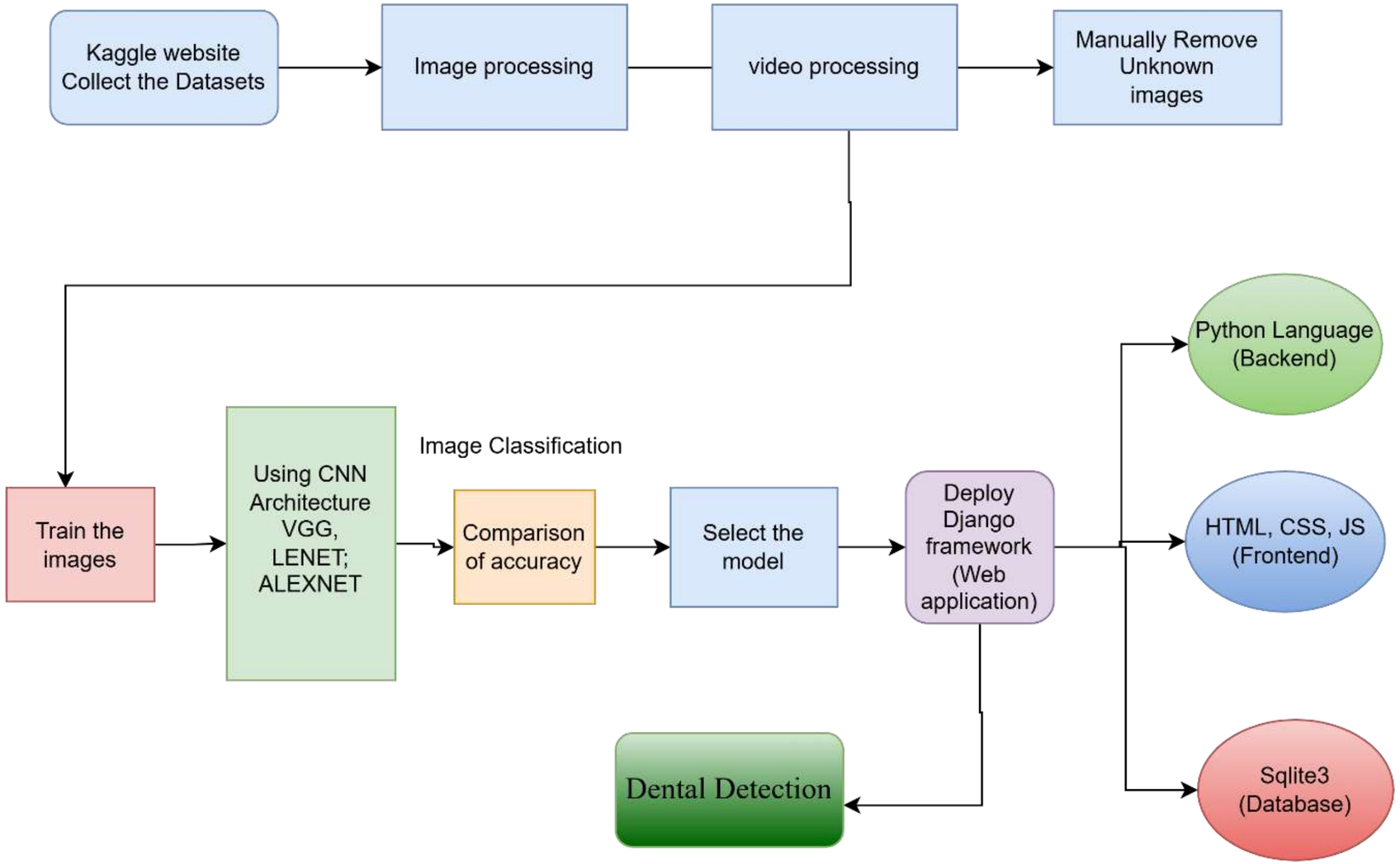
ISSUES AND CHALLENGES

- **Data Quality and Availability :** Crime data is often incomplete and inconsistent, reducing analysis accuracy.
- **Data Bias and Fairness:** Historical data may be biased, leading to unfair crime predictions.
- **Lack of Real-Time Data:** Absence of live data limits instant crime detection and response. Privacy and Security Risks: Sensitive crime data is vulnerable to breaches if not properly protected.
- **Legal and Compliance Challenges:** Strict and varying laws make lawful data usage complex.
- **Data Integration Challenges:** Merging data from multiple agencies and sources is complex and error-prone.
- **Model Interpretability:** Advanced analytical models can be difficult for officers to understand and trust.

PROPOSED WORK

- Deep Learning Model Development: A CNN-based system is designed to automatically detect violence in images and videos. AlexNet and LeNet architectures are implemented for comparative performance evaluation.
- The models are trained using the TensorFlow/Keras deep learning framework.
- Data Processing & Training : Input data undergoes preprocessing such as resizing, normalization, and reshaping.
- Data augmentation techniques improve model generalization and robustness.
- The CNN is trained on datasets containing violent and non-violent samples.
- Real-Time Video Analysis: Uploaded videos are divided into frames for detailed analysis. Each frame is classified using the trained CNN model.
- The system generates an overall violence/non-violence prediction.
- Web Integration & Deployment : The trained model is integrated with a Django-based web application.
  Users can upload images and videos through a user-friendly interface.
- Classification results are displayed instantly for quick decision-making.

ARCHITECTURE DIAGRAM



SYSTEM REQUIREMENTS AND TOOLS

- **Hardware Requirements:**
- Processor: Intel i5 or higher RAM: Minimum 8 GB
- Storage:40 GB Hard Disk or SSD
- **Software Requirements:**
- Operating System: Windows / Linux
- Programming Language: Python
- Tool: Jupyter Notebook
- Libraries: NumPy, Pandas, Scikit-learn, Matplotlib ,TensorFlow

MODULES (1/2)

MODULE 1: DATA PREPARATION & MODEL TRAINING

Input**:** Labeled dataset containing Violence and Non-Violence images/videos

Processing**:** Schema organization, dataset splitting (training/testing), image preprocessing (resizing, normalization), frame extraction from videos, data augmentation (rotation, shear, zoom, flipping)

Output**:** Cleaned, augmented, and model-ready dataset + Trained CNN model (AlexNet / LeNet)

Description **:** This module focuses on preparing high-quality input data and developing the deep learning classifier. Raw images and videos are standardized through preprocessing and augmentation to improve robustness. The dataset is divided into training and testing subsets to enable effective learning and evaluation. CNN architectures such as AlexNet and LeNet are implemented, trained using TensorFlow/Keras, and evaluated based on accuracy metrics.The best-performing model is selected for deployment.

MODULE 2: VIOLENCE DETECTION & SYSTEM DEPLOYMENT

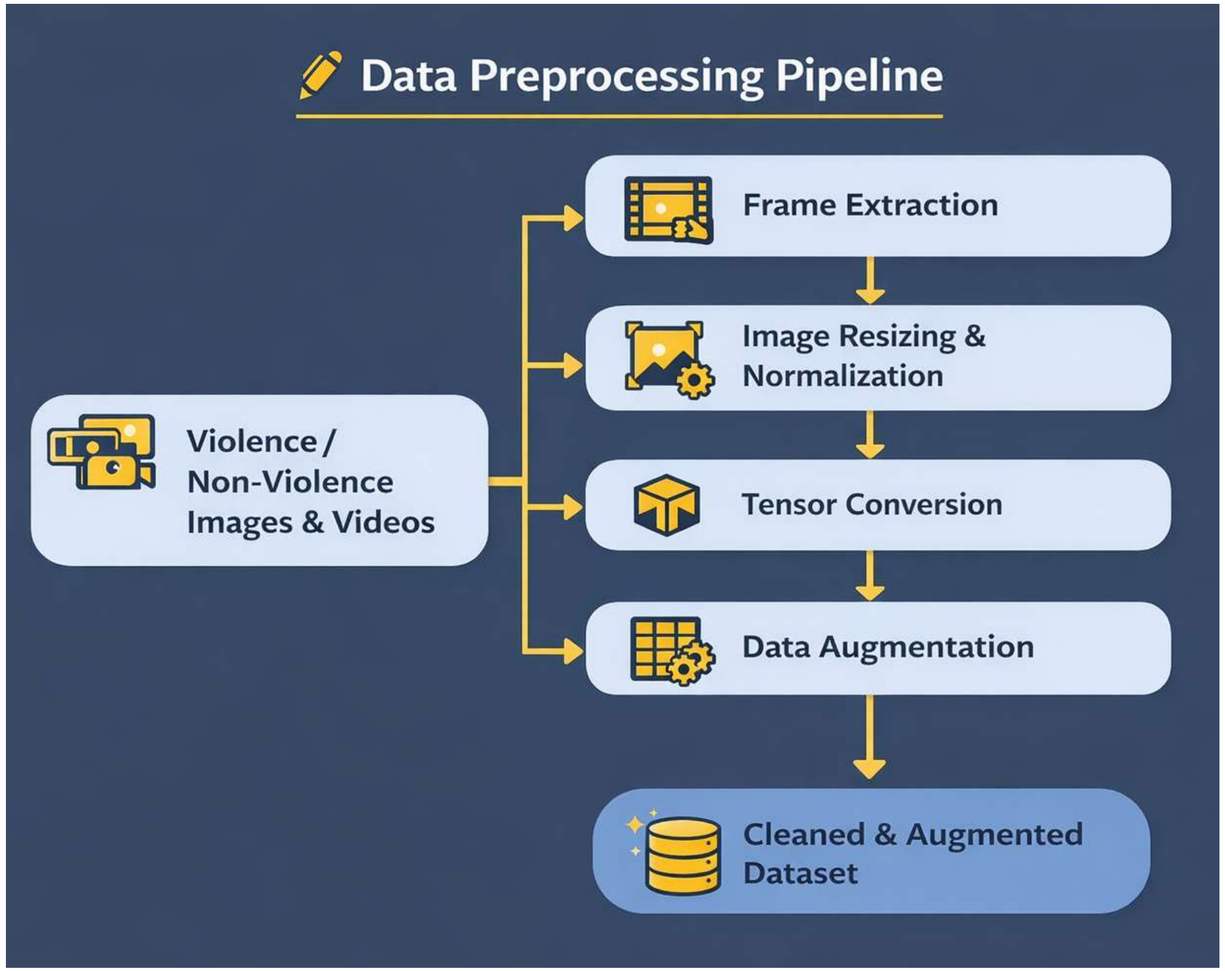
Input**:** User-uploaded images/videos via Django interface

Processing**:** Frame extraction (for videos), preprocessing, CNN-based classification,result aggregation

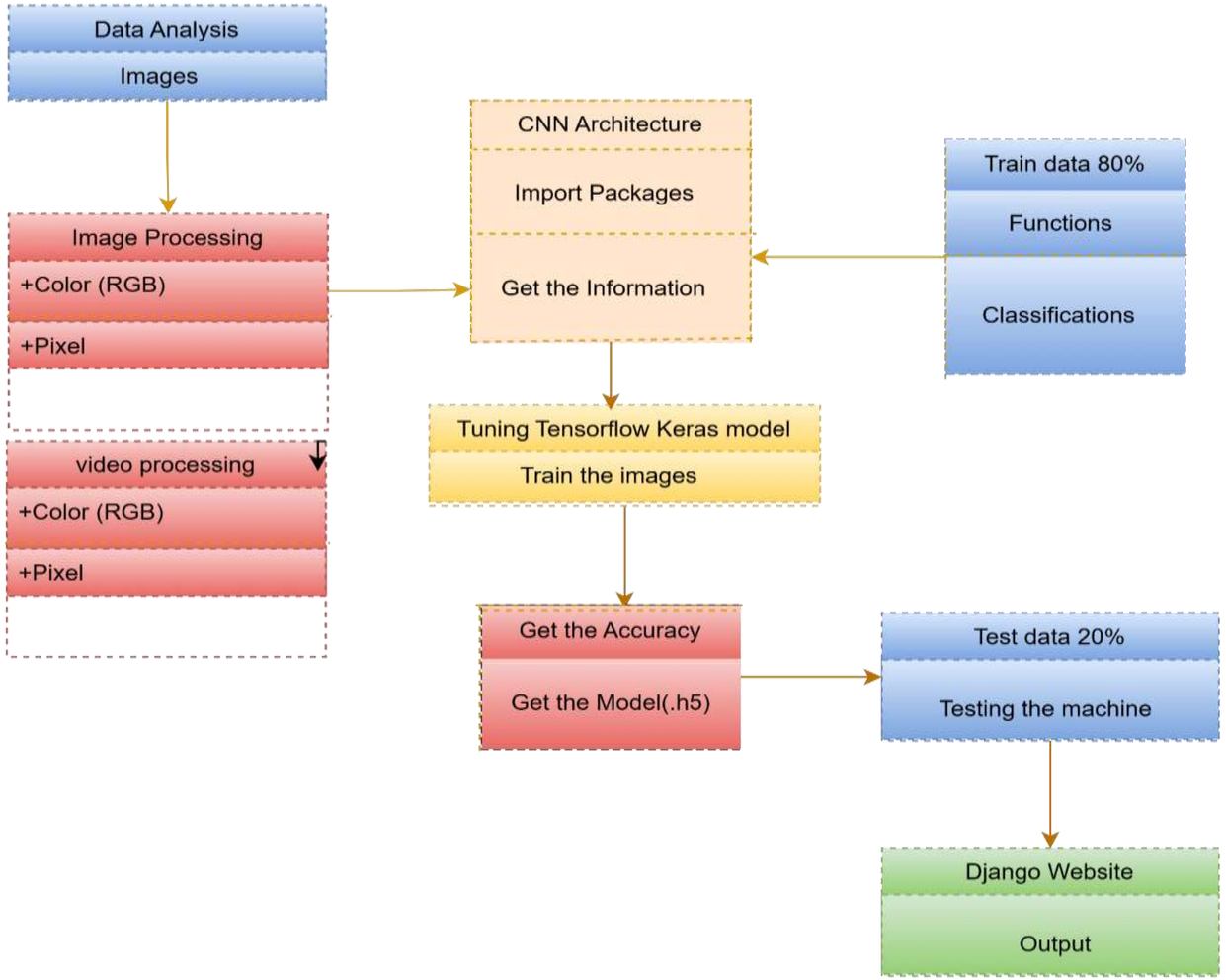
Output**:** Violence / Non-Violence prediction + Visual result display (optional alert trigger)

Description**:** This module applies the trained CNN model in a real-time detection environment. The system accepts images and videos through a user-friendly Django web interface. Uploaded content is processed, and video frames are analyzed individually to detect violent patterns. Classification results are displayed to users, enabling automated monitoring. The module supports surveillance applications, content moderation, and intelligent security systems while ensuring scalability and fast response.

IMPLEMENTATION SCREENSHOTS

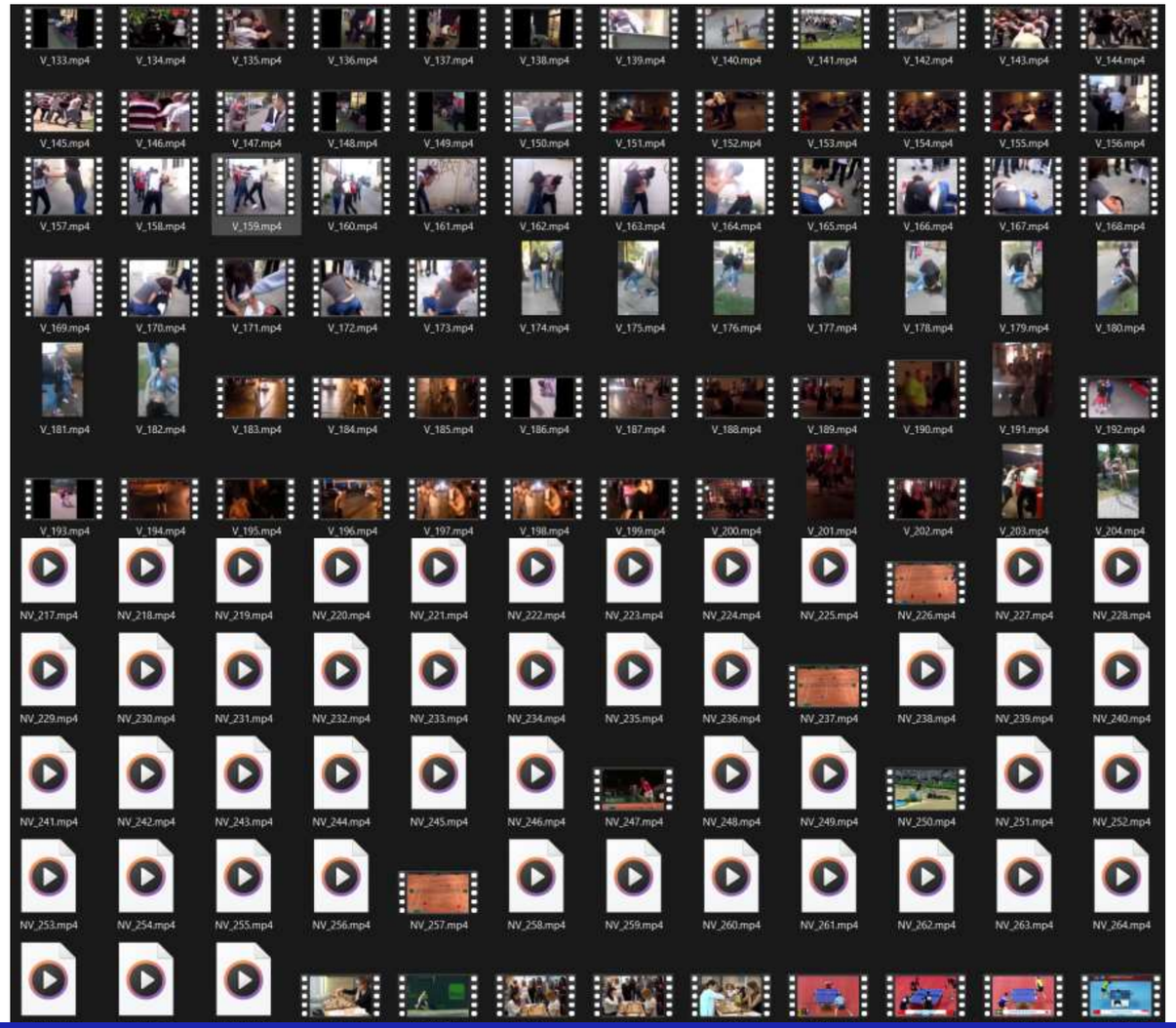


CLASS DIAGRAM



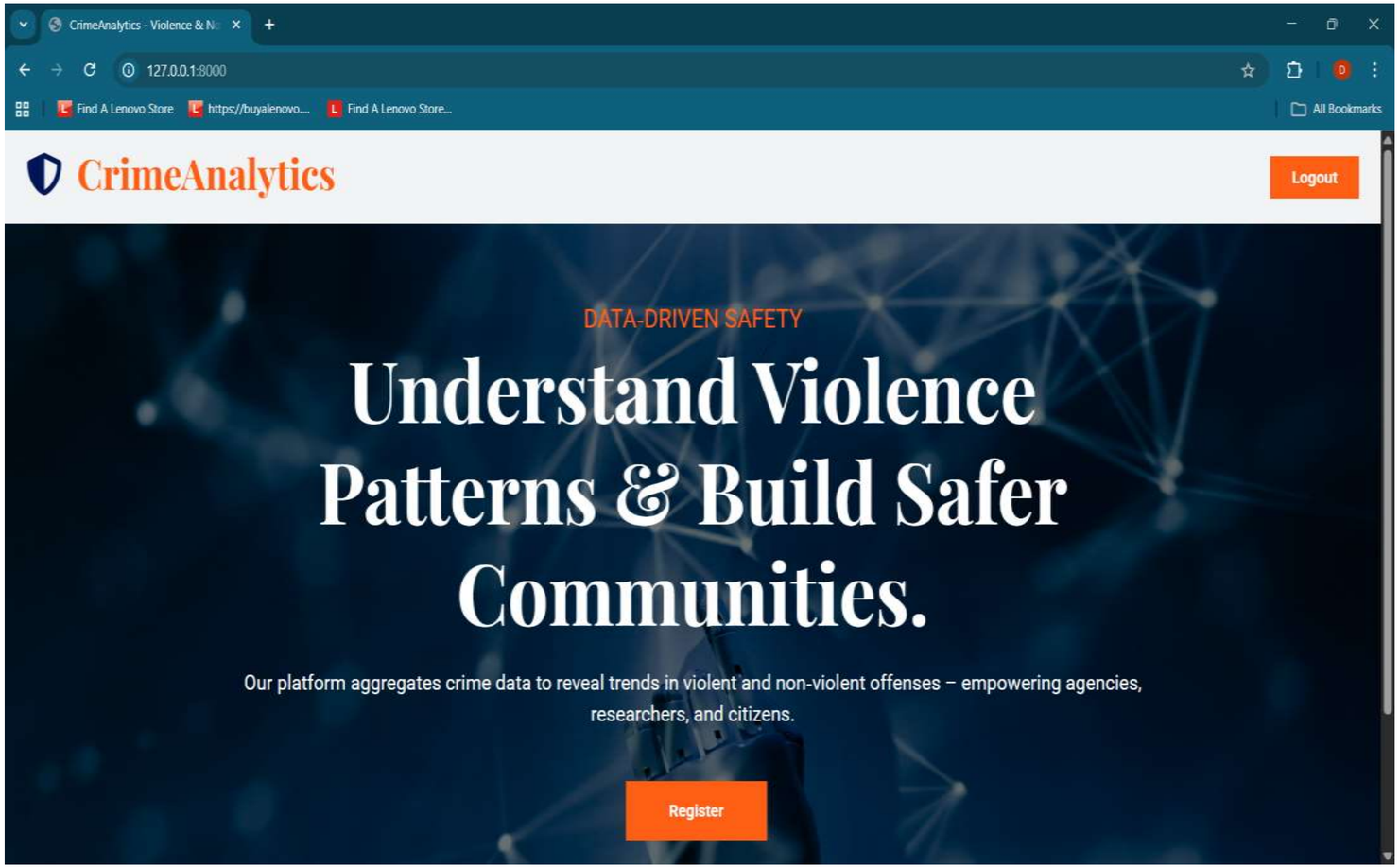
IMPLEMENTATION SCREENSHOTS

VIOLENCE / NON-VIOLENCE DATASETS (VIDEOS AND IMAGES)

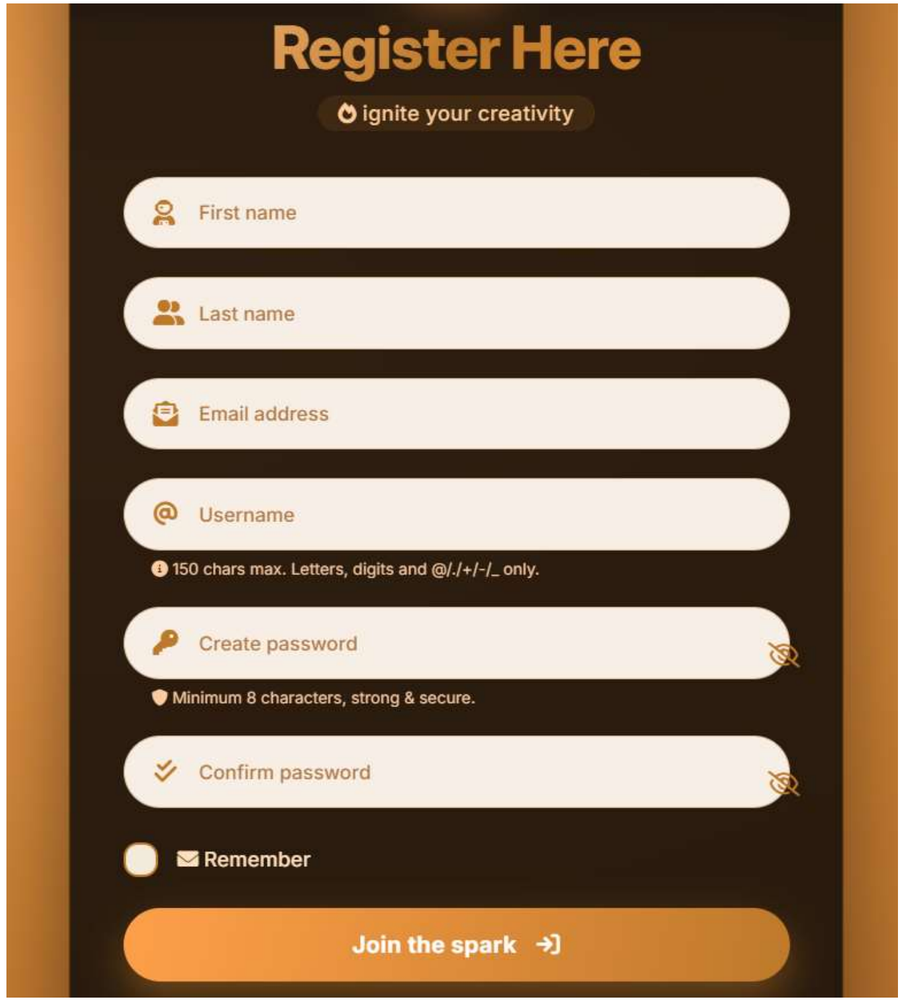


IMPLEMENTATION SCREENSHOTS

Webpage for Crime analytics

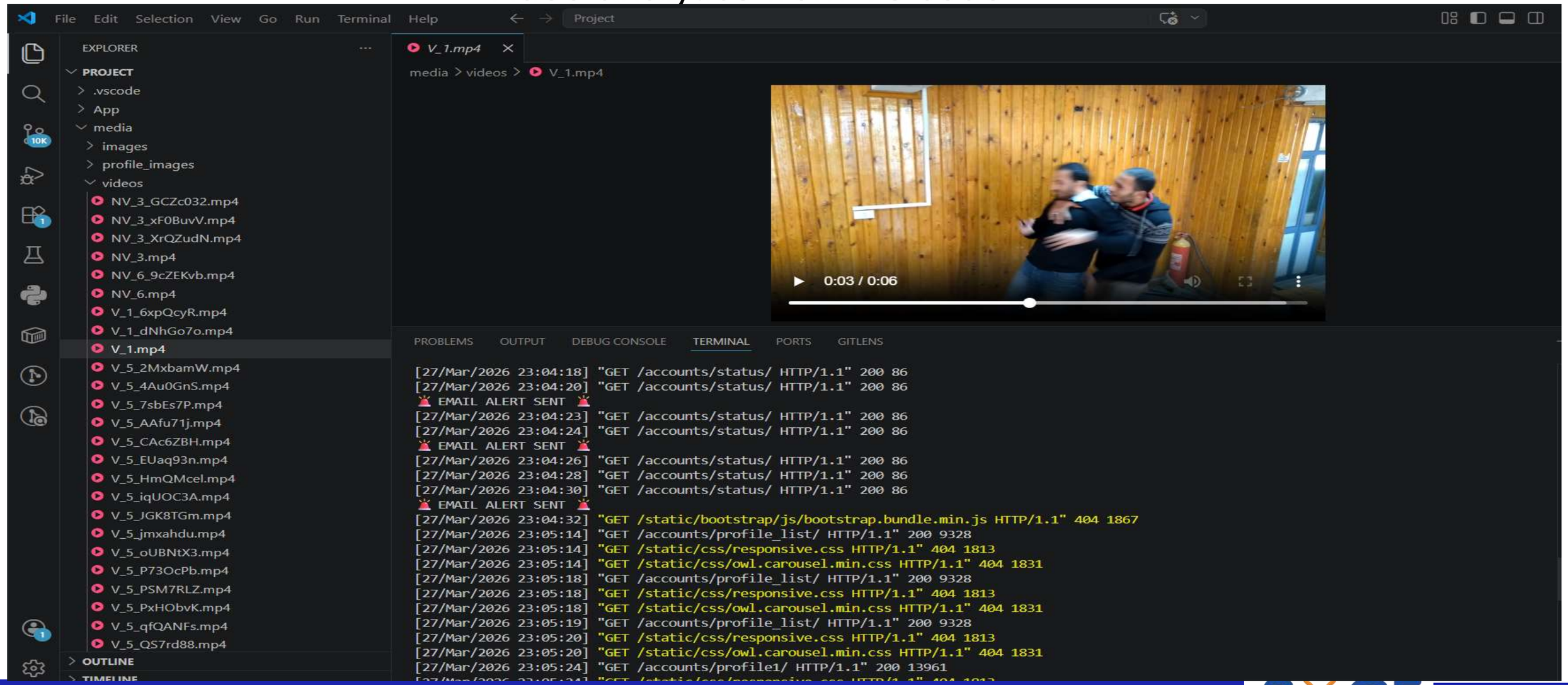


User Login interface



IMPLEMENTATION SCREENSHOTS

Video analytics from Vs code



IMPLEMENTATION SCREENSHOTS

File Edit Selection View Go Run Terminal Help Project

EXPLORER

- PROJECT
  - App
    - static
    - templates
    - __init__.py
    - admin.py
    - apps.py
    - CustomCNN.h5
    - forms.py
    - keras_model.h5
    - Model.h5
    - models.py
    - preview.mp3
    - tests.py
    - urls.py
    - views.py
  - media
    - profile_images
      - default.jpg
      - Folding_marks_01_3.jpg
  - Project
    - __pycache__
    - __init__.py
    - asgi.py
    - settings.py
    - urls.py
    - wsgi.py
  - static
  - db.sqlite3
  - default.jpg
  - manage.py
- OUTLINE
- TIMELINE

manage.py > ...

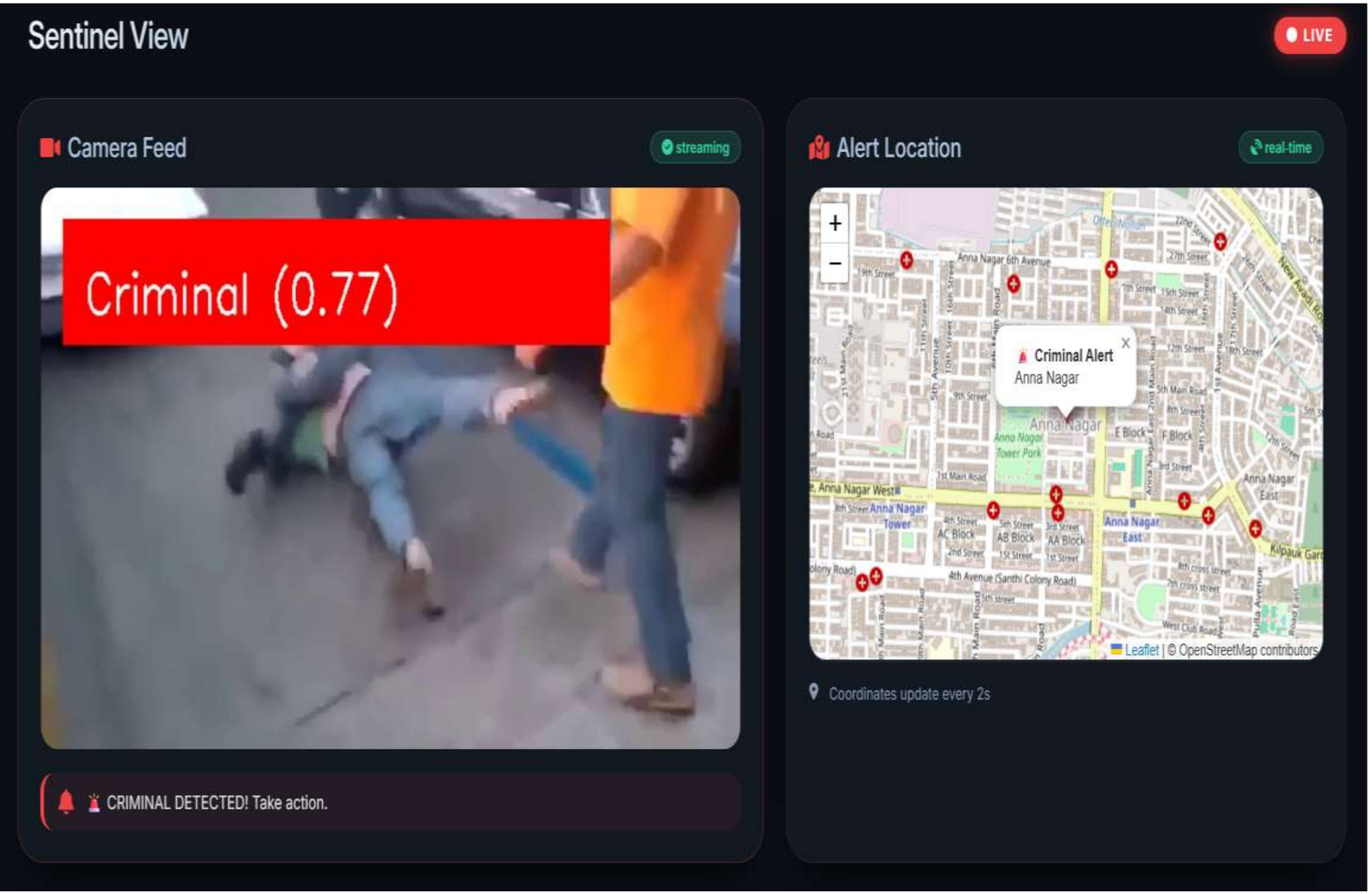
```python
#!/usr/bin/env python
"""Django's command-line utility for administrative tasks."""
import os
import sys


def main():
    """Run administrative tasks."""
    os.environ.setdefault("DJANGO_SETTINGS_MODULE", "Project.settings")
    try:
        from django.core.management import execute_from_command_line
    except ImportError as exc:
        raise ImportError(
            "Couldn't import Django. Are you sure it's installed and "
            "available on your PYTHONPATH environment variable? Did you "
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS

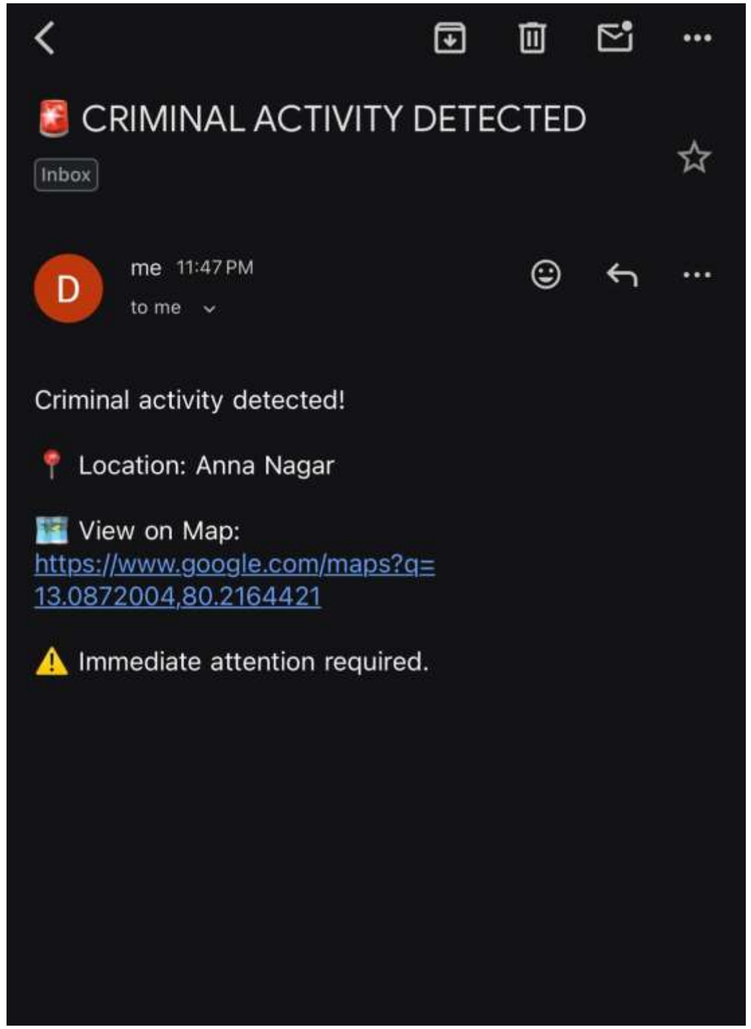
```
tions is deprecated. Please use tf.compat.v1.executing_eagerly_outside_functions instead.

WARNING:tensorflow:From C:\Users\Dharineesh\AN\envs\env\lib\site-packages\keras\src\layers\pooling\max_pooling2d.py:161: The name tf.nn.max_poo
l is deprecated. Please use tf.nn.max_pool2d instead.

Watching for file changes with StatReloader
Performing system checks...

System check identified no issues (0 silenced).
March 27, 2026 - 22:58:06
Django version 4.2.29, using settings 'Project.settings'
Starting development server at http://127.0.0.1:8000/
Quit the server with CTRL-BREAK.

[27/Mar/2026 22:58:25] "GET / HTTP/1.1" 200 6440
[27/Mar/2026 22:59:07] "GET /accounts/register/ HTTP/1.1" 200 25573
[27/Mar/2026 23:00:29] "GET /login/ HTTP/1.1" 200 20822
[27/Mar/2026 23:00:40] "GET /accounts/register/ HTTP/1.1" 200 25573
[27/Mar/2026 23:01:50] "POST /accounts/register/ HTTP/1.1" 200 25838
[27/Mar/2026 23:02:26] "POST /accounts/register/ HTTP/1.1" 302 0
[27/Mar/2026 23:02:26] "GET /login/ HTTP/1.1" 200 20822
[27/Mar/2026 23:02:39] "POST /login/ HTTP/1.1" 302 0
```

IMPLEMENTATION SCREENSHOTS

Crime Alert dashboard



Alert Email to Admin's email



REFERENCES

BASE PAPER:

E.G.Ilgün and M.Dener,“Exploratory data analysis, time series analysis, crime type prediction, and trend forecasting in crime data using machine learning,deep learning, and statistical methods,” *Neural Computing and Applications*, vol.37,pp.11773–11798,2025, doi: 10.1007/s00521-025-11094-9.

REFERENCE PAPER:

- The study performs comprehensive exploratory data analysis (EDA) on large-scale crime datasets from San Francisco, Chicago, and Philadelphia to identify temporal, spatial, and crime-type patterns influencing criminal activities.
- Multiple machine learning models (XGBoost, CatBoost, Random Forest, Decision Tree, MLP, KNN, GNB, and Logistic Regression) are implemented for crime-type prediction, achieving high accuracy, with XGBoost showing the best overall performance.
- Deep learning models (LSTM and BLSTM) and statistical time-series techniques (SARIMA, Prophet, and Holt–Winters) are employed for crime trend forecasting, enabling accurate prediction of future crime hotspots and long-term trends.
- The results demonstrate that integrating machine learning, deep learning, and statistical methods improves predictive policing by supporting optimal police resource allocation and proactive crime prevention strategies.